ShadowBench

The nine-surface shadow AI agent discovery benchmark. Results and tool grades, run of 2026-10-11.

ShadowBench asks one question on every surface where shadow AI agents leave evidence: given what an operator can collect from that surface, which AI agents and which plain AI uses are present, and can a discovery tool find them without flagging the things that merely sound like AI? This run covered 346 cases on nine surfaces (code, identity, gateway, low-code, SaaS, cloud, endpoint, network, runtime) carrying 738 expected entities and 735 decoys. Nine tools ran; four more could not be exercised offline.

Read this first. The rubrics, corpus and adapters are author-written in ShadowScan's own repository and are not independent review. Eight surfaces are synthetic. Per-surface samples are small and the intervals are wide. Four listed tools could not be exercised offline. The results show how each tool behaves on these inputs. They do not estimate field precision or recall, and any ShadowScan lead is suspect until someone without a stake reruns the benchmark. Corpus SHA-256 5997346ad000a0639709718b66743bb794d5f53faff3c2f38e78dbcc8ba36663; generator digest 9a825de5273a7fa8e42c26f2c224e19ee5e86e7f88cf5b7f2bb5e5b47519241e; platform Linux-6.18.44-fc-v114-x86_64-with-glibc2.39.

1. Ranking and grades

Two grades per tool. The in-scope grade weighs entity-level F1, precision discipline (specificity on decoys and raw false positives) and agent-tier discrimination on the surfaces the tool ran. The program grade weighs the same qualities by estate recall (coverage of all expected entities across nine surfaces) and reliability (errors, self-declared incompleteness). Grades are the benchmark author's judgement over the measured numbers, not a formula; the numbers are in metrics.json.

Rank	Tool	Version	Surfaces run	Estate recall	In-scope P / R / F1	Specificity on decoys	Agent recall	In-scope grade	Program grade
1	shadowscan	0.1.2+2d43f508fc95	9/9: cloud, code, endpoint, gateway, identity, lowcode, network, runtime, saas	84% (622/738)	78% / 88% / 83% (80–85)	75%	76% (269/356)	B+	A-*
2	ai-infra-guard	v4.6.5	1/9: network	4% (26/738)	100% / 81% / 90% (84–96)	100%	65% (11/17)	A-	B
3	fleet-ai-tools	fleet ai_tools (fleetdm/fleet main cf12…	2/9: endpoint, runtime	2% (15/738)	88% / 21% / 34% (20–48)	100%	26% (11/43)	B-	C+
4	snyk-agent-scan	0.6.8	1/9: endpoint	1% (7/738)	100% / 29% / 45% (21–65)	n/a	39% (7/18)	B-	C
5	agentmesh-discovery	agentmesh-discovery 5.0.0	3/9: code, endpoint, runtime	2% (15/738)	83% / 8% / 15% (8–23)	100%	14% (15/108)	C	C-
6	vet	v1.20.0	2/9: code, endpoint	2% (18/738)	32% / 14% / 19% (11–28)	n/a	22% (18/83)	C-	D+
7	numbat	dev+77ece3a76dce	1/9: endpoint	0% (3/738)	100% / 12% / 22% (0–44)	n/a	17% (3/18)	C-	D
8	agentdiscover	2.9.5	2/9: code, endpoint	4% (26/738)	10% / 20% / 13% (7–23)	n/a	10% (8/83)	D	D
9	cisco-mcp-scanner	cisco-ai-mcp-scanner 4.8.6	1/9: endpoint	0% (1/738)	100% / 4% / 8% (0–25)	n/a	6% (1/18)	D	D-

- shadowscan (B+ / A-*): The only tool on all nine surfaces, zero crashes; strong on gateway, code, network, low-code and endpoint. Loses the A on precision: identity specificity 57% and SaaS specificity 33%, because it inventories admin-scoped service identities and non-AI workspace bots as findings and calls classic topic-only Copilot Studio bots agents. Twenty live-runtime scans were self-declared incomplete. The asterisk is the conflict of interest: the maintainer wrote the rubrics.
- ai-infra-guard (A- / B): Perfect precision and 100% specificity across the decoy services with 81% recall (it missed Flowise). Its AI-Agent-Config fingerprint fires on any JSON 404 body, so vLLM, TGI and LiteLLM were labelled agent configuration. One surface only.
- fleet-ai-tools (B- / C+): Best endpoint recall of the endpoint-only tools with two false positives, and clean MCP detection on live processes. It does not recognise coding-agent CLIs or model servers from command lines alone; its fleet-scale strength cannot show on a single host.

- snyk-agent-scan (B- / C): No false positives and 39% agent recall. It looks only for MCP configurations and skills, so it never sees Ollama, Aider, Continue, Goose or Cline.
- agentmesh-discovery (C / C-): High precision but 8% recall: a short list of configuration filenames, and process regexes that miss the common @modelcontextprotocol paths. Public preview behaviour.
- vet (C- / D+): Code precision 100% at 8% recall, but it reported plain VS Code as a coding agent on every home with a .vscode directory: 39 false positives in 40 homes.
- numbat (C- / D): Found 3 of 24 endpoint artifacts with no false positives. Its real strength, forensic session reconstruction, is not measured here.
- agentdiscover (D / D): 247 false-positive facts on 26 repositories and 1 of 24 endpoint artifacts found; its configuration paths are macOS-shaped.
- cisco-mcp-scanner (D / D-): 1 of 24 artifacts at 4.4 seconds per home; its Linux path table covers three clients. It is an MCP auditor rather than a discovery tool.

Not graded (no offline entry point): Cartography: needs Neo4j and live cloud APIs; Microsoft Copilot Studio Kit (Agent Inventory): a Power Platform solution; no offline entry point; Google k8s-aibom: an in-cluster controller with no manifest mode; AgentSight: eBPF uprobes on live TLS libraries.

2. Headline metrics

Estate recall counts every expected entity in the corpus, charging a tool for the surfaces it does not cover. In-scope metrics count only the surfaces and modalities a tool ran on. Specificity is measured on decoy entities. Agent recall is the share of expected agents a tool reported as agents; agent false alarms are plain-AI entities and decoys it called agents. F1 intervals are 2,000-sample case bootstraps. An error (crash, timeout, unreadable output) counts every expected entity as missed; an incomplete run is one the tool itself marked incomplete.

Tool	Surfaces	Estate recall	In-scope P / R / F1	TP / FP / FN / TN	Specificity	Agent recall	Agent false alarms	Case balanced acc.	Errors	Incomplete	Seconds med / p90
shadowscan	9/9	84%	78% / 88% / 83% (80–85)	622 / 176 / 84 / 512	75%	76% (269/356)	16% (164/1035)	80%	0	27	1.8 / 3.0
ai-infra-guard	1/9	4%	100% / 81% / 90% (84–96)	26 / 0 / 6 / 50	100%	65% (11/17)	17% (11/65)	100%	0	0	1.2 / 4.9
fleet-ai-tools	2/9	2%	88% / 21% / 34% (20–48)	15 / 2 / 57 / 81	100%	26% (11/43)	0% (0/110)	64%	0	0	0.0 / 0.0
snyk-agent-scan	1/9	1%	100% / 29% / 45% (21–65)	7 / 0 / 17 / 0	n/a	39% (7/18)	0% (0/6)	65%	0	0	1.0 / 1.1
agentmesh-discovery	3/9	2%	83% / 8% / 15% (8–23)	15 / 3 / 166 / 81	100%	14% (15/108)	0% (0/154)	61%	0	0	0.3 / 0.3
vet	2/9	2%	32% / 14% / 19% (11–28)	18 / 39 / 115 / 0	n/a	22% (18/83)	0% (0/50)	57%	0	0	0.2 / 0.2
numbat	1/9	0%	100% / 12% / 22% (0–44)	3 / 0 / 21 / 0	n/a	17% (3/18)	0% (0/6)	56%	0	0	0.0 / 0.0
agentdiscover	2/9	4%	10% / 20% / 13% (7–23)	26 / 247 / 107 / 0	n/a	10% (8/83)	0% (0/50)	62%	0	0	1.3 / 3.8
cisco-mcp-scanner	1/9	0%	100% / 4% / 8% (0–25)	1 / 0 / 23 / 0	n/a	6% (1/18)	0% (0/6)	52%	0	0	4.1 / 5.0

3. Results by surface

code: 26 cases, 16 positive; 109 expected entities, 0 decoys; modalities checkout

Tool	Modalities	P / R / F1	TP / FP / FN / TN	Specificity	Agent recall	Agent false alarms	Case recall / specificity	Errors	Incomplete	Seconds median
shadowscan	checkout	97% / 91% / 94% (90–96)	99 / 3 / 10 / 0	n/a	95%	0%	100% / 100%	0	7	2.8
agentmesh-discovery	checkout	83% / 5% / 9% (0–20)	5 / 1 / 104 / 0	n/a	8%	0%	25% / 100%	0	0	0.3
vet	checkout	100% / 8% / 15% (6–26)	9 / 0 / 100 / 0	n/a	14%	0%	38% / 100%	0	0	0.1

Tool	Modalities	P / R / F1	TP / FP / FN / TN	Specificity	Agent recall	Agent false alarms	Case recall / specificity	Errors	Incomplete	Seconds median
agentdiscover	checkout	9% / 23% / 13% (7–25)	25 / 247 / 84 / 0	n/a	11%	0%	62% / 90%	0	0	1.4

identity: 40 cases, 24 positive; 96 expected entities, 159 decoys; modalities entra-export, okta-export

Tool	Modalities	P / R / F1	TP / FP / FN / TN	Specificity	Agent recall	Agent false alarms	Case recall / specificity	Errors	Incomplete	Seconds median
shadowscan	entra-export, okta-export	57% / 97% / 72% (63–79)	93 / 69 / 3 / 90	57%	47%	23%	100% / 0%	0	0	1.9

gateway: 40 cases, 40 positive; 124 expected entities, 0 decoys; modalities bedrock-export, litellm-export

Tool	Modalities	P / R / F1	TP / FP / FN / TN	Specificity	Agent recall	Agent false alarms	Case recall / specificity	Errors	Incomplete	Seconds median
shadowscan	bedrock-export, litellm-export	100% / 100% / 100% (100–100)	124 / 0 / 0 / 0	n/a	100%	0%	100% / n/a	0	0	2.7

lowcode: 40 cases, 24 positive; 78 expected entities, 118 decoys; modalities n8n-export, powerplatform-export

Tool	Modalities	P / R / F1	TP / FP / FN / TN	Specificity	Agent recall	Agent false alarms	Case recall / specificity	Errors	Incomplete	Seconds median
shadowscan	n8n-export, powerplatform-export	88% / 100% / 93% (87–98)	78 / 11 / 0 / 107	91%	100%	27%	100% / 75%	0	0	1.7

saas: 40 cases, 24 positive; 88 expected entities, 110 decoys; modalities github-export, slack-export

Tool	Modalities	P / R / F1	TP / FP / FN / TN	Specificity	Agent recall	Agent false alarms	Case recall / specificity	Errors	Incomplete	Seconds median
shadowscan	github-export, slack-export	51% / 88% / 64% (53–72)	77 / 74 / 11 / 36	33%	66%	24%	100% / 6%	0	0	1.8

cloud: 40 cases, 24 positive; 77 expected entities, 133 decoys; modalities aws-export, k8s-export

Tool	Modalities	P / R / F1	TP / FP / FN / TN	Specificity	Agent recall	Agent false alarms	Case recall / specificity	Errors	Incomplete	Seconds median
shadowscan	aws-export, k8s-export	97% / 75% / 85% (78–90)	58 / 2 / 19 / 131	98%	41%	11%	96% / 94%	0	0	1.7

endpoint: 40 cases, 24 positive; 24 expected entities, 0 decoys; modalities home

Tool	Modalities	P / R / F1	TP / FP / FN / TN	Specificity	Agent recall	Agent false alarms	Case recall / specificity	Errors	Incomplete	Seconds median
shadowscan	home	100% / 83% / 91% (80–98)	20 / 0 / 4 / 0	n/a	94%	0%	83% / 100%	0	0	1.7
fleet-ai-tools	home	83% / 42% / 56% (34–74)	10 / 2 / 14 / 0	n/a	33%	0%	42% / 88%	0	0	0.0
snyk-agent-scan	home	100% / 29% / 45% (21–65)	7 / 0 / 17 / 0	n/a	39%	0%	29% / 100%	0	0	1.0
agentmesh-discovery	home	67% / 17% / 27% (7–47)	4 / 2 / 20 / 0	n/a	22%	0%	17% / 88%	0	0	0.2
vet	home	19% / 38% / 25% (13–37)	9 / 39 / 15 / 0	n/a	50%	0%	100% / 0%	0	0	0.2
numbat	home	100% / 12% / 22% (0–44)	3 / 0 / 21 / 0	n/a	17%	0%	12% / 100%	0	0	0.0
agentdiscover	home	100% / 4% / 8% (0–25)	1 / 0 / 23 / 0	n/a	6%	0%	4% / 100%	0	0	1.3
cisco-mcp-scanner	home	100% / 4% / 8% (0–25)	1 / 0 / 23 / 0	n/a	6%	0%	4% / 100%	0	0	4.1

network: 40 cases, 24 positive; 44 expected entities, 50 decoys; modalities active, passive

Tool	Modalities	P / R / F1	TP / FP / FN / TN	Specificity	Agent recall	Agent false alarms	Case recall / specificity	Errors	Incomplete	Seconds median
shadowscan	passive	100% / 92% / 96% (84–100)	11 / 0 / 1 / 0	n/a	89%	0%	92% / 100%	0	0	1.7
ai-infra-guard	active	100% / 81% / 90% (84–96)	26 / 0 / 6 / 50	100%	65%	17%	100% / 100%	0	0	1.2

runtime: 40 cases, 24 positive; 98 expected entities, 165 decoys; modalities live, snapshot

Tool	Modalities	P / R / F1	TP / FP / FN / TN	Specificity	Agent recall	Agent false alarms	Case recall / specificity	Errors	Incomplete	Seconds median
shadowscan	live, snapshot	78% / 63% / 70% (63–76)	62 / 17 / 36 / 148	90%	77%	11%	100% / 56%	0	20	1.8
fleet-ai-tools	live	100% / 10% / 19% (0–34)	5 / 0 / 43 / 81	100%	20%	0%	25% / 100%	0	0	0.0
agentmesh-discovery	live	100% / 12% / 22% (11–32)	6 / 0 / 42 / 81	100%	24%	0%	50% / 100%	0	0	0.3

4. Method

Cases and truth. A case is one input a tool is pointed at: an identity-provider export (Okta apps with grants and tokens; Graph service principals, applications, permission grants and app-role assignments), a gateway log window (LiteLLM spend logs; Bedrock model invocation logs), a low-code export (Power Platform flows, apps, Copilot Studio bots and components; n8n workflows), a SaaS admin export (Slack approved apps, requests, users; GitHub App installations), a cloud account (AWS CLI outputs for STS, Bedrock Agents, AgentCore, Lambda, SageMaker, IAM; a Kubernetes workload list), a developer home directory, a workstation's Zeek logs or a set of loopback services, a host's osquery process table or live inert processes, or a pinned repository checkout. Every case lists expected entities with a label (agent or llm) and the evidence behind it, tolerated keys a tool may report without penalty, and decoys labeled none.

Labels. agent: an autonomous or tool-using AI agent or an agent integration (a configured MCP server, Copilot Studio agent, Bedrock agent, coding-agent CLI, agent framework process). llm: AI use without agent evidence (a chat SaaS app, a model server, a single-completion caller, a GPT step in a flow). none: no AI, including service accounts without AI evidence, infrastructure agents, Minecraft Bedrock servers, apps named copilot, classic chatbots and plain workloads.

Matching. A reported key matches an expected or decoy key when equal or, for home-directory artifacts, when one path is a parent directory of the other. Reported keys are de-duplicated. A key matching nothing expected or tolerated is a false positive; one hitting a decoy also counts against specificity. Code cases use the fact

vocabulary of the code-surface benchmark (category:value), with provider facts labeled llm and all others agent.

Isolation. Every tool runs as root in fresh network and PID namespaces with a minimal environment, a case-local HOME and no inherited credentials. Active-network cases serve mock products on loopback inside the tool's namespace; live-runtime cases spawn an inert static binary under each recorded process name and command line. No tool is given an allowlist, an inventory or a known-agent list. Verbatim tool output is kept beside the normalized entities.

Corpus construction. Export-style cases draw entity families by quota (45% agent, 15% plain AI, 40% decoy, at least one decoy per export), redraw collisions and render the platform's own file shapes. Endpoint and passive-network cases reuse the pre-registered families of the earlier head-to-head benchmark. Code cases are 26 public repositories pinned by commit (16 positive, 4 control, 6 near-miss). The corpus records its seed and a SHA-256 over the generator sources; verify --regenerate rebuilds it byte for byte.

Bias controls. Rubrics, families, generator, adapters and scoring were committed before the scored run; adapters were calibrated on probe inputs and a separate four-case-per-surface smoke corpus. Decoys come from the confusions real tools make. Tools are scored only on surfaces they declare; estate recall shows the cost of uncovered surfaces. A tool's own notion of a finding is used, with no confidence threshold imposed on top.

Limits. Eight surfaces are synthetic and author-written; the code surface's labels are session-labeled evidence, not independent review. The same maintainer wrote ShadowScan, the rubrics and every adapter. Live-only tools without an offline entry point were not run. Mock services reproduce banners and version endpoints, not full applications. The gateway surface has no decoys. Twenty cases per modality give wide intervals.

5. Reproduction

Repository: github.com/aisecnomad/Project-Nexus, benchmarks/shadowbench and tools/shadowbench. Corpus SHA-256 5997346ad000a0639709718b66743bb794d5f53faff3c2f38e78dbcc8ba36663. Generator digest 9a825de5273a7fa8e42c26f2c224e19ee5e86e7f88cf5b7f2bb5e5b47519241e. Run started 2026-10-11T18:10:09+00:00, finished 2026-10-11T18:18:56+00:00, 3 parallel workers on Linux-6.18.44-fc-v114-x86_64-with-glibc2.39, Python 3.13.16.

Tool	Version or commit benchmarked
shadowscan	0.1.2+2d43f508fc95
ai-infra-guard	v4.6.5
fleet-ai-tools	fleet ai_tools (fleetdm/fleet main cf1217bafcd5cdce367165b1c604f94ed559e3d0, 2026-10-11; upstream karmine05/agentic-detector v0.3.0 7c942d0)
snyk-agent-scan	0.6.8
agentmesh-discovery	agentmesh-discovery 5.0.0
vet	v1.20.0
numbat	dev+77ece3a76dce
agentdiscover	2.9.5
cisco-mcp-scanner	cisco-ai-mcp-scanner 4.8.6

Commands: python -m tools.shadowbench verify --corpus benchmarks/shadowbench/corpus.json --regenerate; bash tools/shadowbench/install_tools.sh TOOL_ROOT; python tools/shadowbench/fetch_code.py CHECKOUTS; python -m tools.shadowbench run --corpus ... --tools tools.json --checkouts CHECKOUTS --out RESULTS; python -m tools.shadowbench score ...; python -m tools.shadowbench report ...; python -m tools.shadowbench pdf ....